STAT3888: Ordinal Logistic Regression

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In this document, we will try to use ordinal logistic regression instead of our previous multinomial logistic regression model, as `ALT3` and `GGT3` both have clear orders in their categories (Low < Borderline < High).

▼ Code

```
# Initialise data
pdat.full = readRDS(here("Objects", "pdat_full.rds"))
fdat.full = readRDS(here("Objects", "fdat_full.rds"))

# Order ALT3 and GGT3
pdat.full = pdat.full %>%
  mutate(ALT3 = factor(ALT3, c("Low", "Borderline", "High"), orc
         GGT3 = factor(GGT3, c("Low", "Borderline", "High"), orc

fdat.full = fdat.full %>%
  mutate(ALT3 = factor(ALT3, c("Low", "Borderline", "High"), orc
         GGT3 = factor(GGT3, c("Low", "Borderline", "High"), orc
```

Model Fitting

We will use the `ordinal::clm()` function to fit the models, and then using `ordinal::nominal_test()` for the POA hypothesis test. We can also use the `MASS::polr()` function to see whether there are differences or not.

▼ Code

```
# Fitting ALT and GGT model under PDAT
alt.olr = clm(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB, d
ggt.olr = clm(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB, d

alt.olr
```

```
formula: ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB
data:    pdat.full

 link  threshold nobs logLik  AIC    niter max.grad cond.H
 logit flexible  460  -315.23 674.46 18(0) 5.47e-09 3.9e+10
```

```
Coefficients:
EXREGUICDid not meet recommended guidelines
                                   -0.335459
                              AGEB25-29 years
                                   -2.512454
                              AGEB30-34 years
                                   -2.703939
                              AGEB35-39 years
                                   -3.573217
                              AGEB40-44 years
                                   -1.932144
                              AGEB45-49 years
                                   -2.011342
                              AGEB50-54 years
                                   -2.590332
                              AGEB55-59 years
                                   -2.020760
                              AGEB60-64 years
                                   -1.393202
                              AGEB65-69 years
                                   -1.668019
                              AGEB70-74 years
                                   -3.025362
                              AGEB75-79 years
                                   -3.488726
                              AGEB80-84 years
                                   -3.138411
                        AGEB85 years and older
                                  -22.245417
                                     SEXMale
                                    1.099836
                               WAISTNot at risk
                                   -1.289520
             WAISTSubstantially increased risk
                                    0.254824
                                DSMKHHRBOne
                                    0.009778
                                DSMKHHRBTwo
                                    0.539508
                         DSMKHHRBThree or more
                                    0.951036

Threshold coefficients:
 Low|Borderline Borderline|High
       -0.88461        -0.09551
(902 observations deleted due to missingness)
```

▼ Code

```
ggt.olr
```

```
formula: GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB
data:    pdat.full

 link  threshold nobs logLik  AIC    niter max.grad cond.H
 logit flexible  460  -361.16 766.32 20(1) 4.07e-09 2.5e+12

Coefficients:
EXREGUICDid not meet recommended guidelines
                                    0.12649
                            AGEB25-29 years
                                   19.59726
                            AGEB30-34 years
                                   19.62580
                            AGEB35-39 years
                                   19.33826
                            AGEB40-44 years
                                   20.48147
                            AGEB45-49 years
                                   20.41281
                            AGEB50-54 years
                                   20.65401
                            AGEB55-59 years
                                   20.20772
                            AGEB60-64 years
                                   21.06597
                            AGEB65-69 years
                                   21.07894
                            AGEB70-74 years
                                   20.48091
                            AGEB75-79 years
                                   20.31776
                            AGEB80-84 years
                                   20.47438
                  AGEB85 years and older
                                   20.60114
                                    SEXMale
                                    0.59049
                           WAISTNot at risk
                                   -0.09379
          WAISTSubstantially increased risk
                                    0.49185
                               DSMKHHRBOne
                                    0.12755
```

```
                                    DSMKHHRBTwo
                                        0.71546
                              DSMKHHRBThree or more
                                      -20.49470

Threshold coefficients:
 Low|Borderline Borderline|High
          21.78           22.25
(902 observations deleted due to missingness)
```

▼ Code

```
# MASS implementation for ALT and GGT under PDAT
alt.polr = polr(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
ggt.polr = polr(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,

alt.polr
```

```
Call:
polr(formula = ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
    data = pdat.full)

Coefficients:
EXREGUICDid not meet recommended guidelines
                                -0.335429358
                            AGEB20-24 years
                                17.436820940
                            AGEB25-29 years
                                14.924444789
                            AGEB30-34 years
                                14.732914119
                            AGEB35-39 years
                                13.863609373
                            AGEB40-44 years
                                15.504920708
                            AGEB45-49 years
                                15.425590967
                            AGEB50-54 years
                                14.846565613
                            AGEB55-59 years
                                15.416073198
                            AGEB60-64 years
                                16.043724460
                            AGEB65-69 years
                                15.768964546
                            AGEB70-74 years
                                14.411743942
```

```
                              AGEB75-79 years
                                 13.948134683
                              AGEB80-84 years
                                 14.298567367
                                      SEXMale
                                  1.099757727
                             WAISTNot at risk
                                 -1.289457005
          WAISTSubstantially increased risk
                                  0.254830003
                                  DSMKHHRBOne
                                  0.009663606
                                  DSMKHHRBTwo
                                  0.539414806
                       DSMKHHRBThree or more
                                  0.950441128

Intercepts:
 Low|Borderline Borderline|High
       16.55226        17.34131

Residual Deviance: 630.4625
AIC: 674.4625
(902 observations deleted due to missingness)
```

▼ Code

```
ggt.polr
```

```
Call:
polr(formula = GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
    data = pdat.full)

Coefficients:
EXREGUICDid not meet recommended guidelines
                                  0.12649342
                              AGEB20-24 years
                                 -14.90748213
                              AGEB25-29 years
                                  -1.00389483
                              AGEB30-34 years
                                  -0.97538181
                              AGEB35-39 years
                                  -1.26295193
                              AGEB40-44 years
                                  -0.11969125
                              AGEB45-49 years
```

```
                                    -0.18837167
                                AGEB50–54 years
                                     0.05284356
                                AGEB55–59 years
                                    -0.39346069
                                AGEB60–64 years
                                     0.46479175
                                AGEB65–69 years
                                     0.47776194
                                AGEB70–74 years
                                    -0.12024811
                                AGEB75–79 years
                                    -0.28341570
                                AGEB80–84 years
                                    -0.12681963
                                        SEXMale
                                     0.59048494
                               WAISTNot at risk
                                    -0.09378305
                 WAISTSubstantially increased risk
                                     0.49185791
                                DSMKHHRBOne
                                     0.12753975
                                DSMKHHRBTwo
                                     0.71546732
                           DSMKHHRBThree or more
                                   -14.66886031

Intercepts:
 Low|Borderline Borderline|High
       1.180366        1.649777

Residual Deviance: 722.3224
AIC: 766.3224
(902 observations deleted due to missingness)
```

Using `clm()`, we obtain the following warning: “(1) Hessian is numerically singular: parameters are not uniquely determined; In addition: Absolute convergence criterion was met, but relative criterion was not met”. Using `polr()`, we obtain the following warning: “design appears to be rank-deficient, so dropping some coefs”. It seems that these models may not be very stable at least under PDAT, so let us try using FDAT.

▼ Code

```
# clm Models using FDAT
alt.folr = clm(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
```

```
ggt.folr = clm(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,

alt.folr
```

```
formula: ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB
data:    fdat.full

 link  threshold nobs logLik   AIC     niter max.grad cond.H
 logit flexible  5410 -3500.97 7047.93 6(0)  9.27e-09 5.1e+03

Coefficients:
EXREGUICDid not meet recommended guidelines
                                    0.11638
                           AGEB20-24 years
                                    0.01370
                           AGEB25-29 years
                                    0.36326
                           AGEB30-34 years
                                    0.13976
                           AGEB35-39 years
                                    0.27087
                           AGEB40-44 years
                                    0.14180
                           AGEB45-49 years
                                    0.41537
                           AGEB50-54 years
                                    0.42857
                           AGEB55-59 years
                                    0.13954
                           AGEB60-64 years
                                    0.18030
                           AGEB65-69 years
                                   -0.16808
                           AGEB70-74 years
                                   -0.47719
                           AGEB75-79 years
                                   -1.37943
                           AGEB80-84 years
                                   -1.72578
                 AGEB85 years and older
                                   -2.77671
                                    SEXMale
                                    0.85427
                         WAISTNot at risk
                                   -0.66717
          WAISTSubstantially increased risk
                                    0.64970
```

```
                                    DSMKHHRBOne
                                      -0.16645
                                    DSMKHHRBTwo
                                      -0.26535
                           DSMKHHRBThree or more
                                       0.06928

Threshold coefficients:
 Low|Borderline Borderline|High
          1.888           2.743
(10065 observations deleted due to missingness)
```

▼ Code

```
ggt.folr
```

```
formula: GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB
data:    fdat.full

 link  threshold nobs logLik   AIC     niter max.grad cond.H
 logit flexible  5410 -3072.47 6190.94 8(1)  1.81e-12 5.9e+04

Coefficients:
EXREGUICDid not meet recommended guidelines
                                       0.30843
                                AGEB20-24 years
                                       1.43559
                                AGEB25-29 years
                                       1.68377
                                AGEB30-34 years
                                       1.93958
                                AGEB35-39 years
                                       1.91931
                                AGEB40-44 years
                                       2.13671
                                AGEB45-49 years
                                       2.26914
                                AGEB50-54 years
                                       2.54658
                                AGEB55-59 years
                                       2.37664
                                AGEB60-64 years
                                       2.57883
                                AGEB65-69 years
                                       2.48986
                                AGEB70-74 years
                                       2.27990
```

```
                                AGEB75-79 years
                                        2.31923
                                AGEB80-84 years
                                        2.21934
                           AGEB85 years and older
                                        2.32107
                                        SEXMale
                                        0.05955
                              WAISTNot at risk
                                       -0.55847
                  WAISTSubstantially increased risk
                                        0.47847
                                   DSMKHHRBOne
                                        0.27773
                                   DSMKHHRBTwo
                                        0.58242
                            DSMKHHRBThree or more
                                        0.60320

Threshold coefficients:
 Low|Borderline Borderline|High
          4.109           4.546
(10065 observations deleted due to missingness)
```

▼ Code

```
# polr Models using FDAT
alt.fpolr = polr(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRE
                 data = fdat.full, Hess = TRUE)
ggt.fpolr = polr(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRE
                 data = fdat.full, Hess = TRUE)

summary(alt.fpolr)
```

```
Call:
polr(formula = ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
    data = fdat.full, Hess = TRUE)

Coefficients:
                                                Value Std. Error
t value
EXREGUICDid not meet recommended guidelines  0.11638    0.06825
1.7052
AGEB20-24 years                              0.01413    0.42425
0.0333
AGEB25-29 years                              0.36368    0.38874
0.9355
```

```
AGEB30–34 years                              0.14019    0.38939
0.3600
AGEB35–39 years                              0.27129    0.38554
0.7037
AGEB40–44 years                              0.14222    0.38313
0.3712
AGEB45–49 years                              0.41579    0.37937
1.0960
AGEB50–54 years                              0.42899    0.37929
1.1310
AGEB55–59 years                              0.13997    0.38117
0.3672
AGEB60–64 years                              0.18073    0.37912
0.4767
AGEB65–69 years                             -0.16766    0.38487
-0.4356
AGEB70–74 years                             -0.47676    0.39561
-1.2051
AGEB75–79 years                             -1.37899    0.43810
-3.1476
AGEB80–84 years                             -1.72533    0.49468
-3.4878
AGEB85 years and older                      -2.77616    0.80886
-3.4322
SEXMale                                      0.85427    0.06836
12.4960
WAISTNot at risk                            -0.66716    0.10378
-6.4288
WAISTSubstantially increased risk            0.64970    0.08431
7.7061
DSMKHHRBOne                                 -0.16645    0.09697
-1.7165
DSMKHHRBTwo                                 -0.26534    0.18443
-1.4387
DSMKHHRBThree or more                        0.06928    0.55482
0.1249

Intercepts:
               Value   Std. Error t value
Low|Borderline   1.8885  0.3787     4.9867
Borderline|High  2.7430  0.3800     7.2177

Residual Deviance: 7001.932
AIC: 7047.932
(10065 observations deleted due to missingness)
```

▼ Code

```
summary(ggt.fpolr)
```

```
Call:
polr(formula = GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
    data = fdat.full, Hess = TRUE)

Coefficients:
                                                 Value Std. Error
t value
EXREGUICDid not meet recommended guidelines  0.30844    0.07312
4.218
AGEB20-24 years                              1.44968    1.06137
1.366
AGEB25-29 years                              1.69786    1.03441
1.641
AGEB30-34 years                              1.95369    1.03042
1.896
AGEB35-39 years                              1.93333    1.02829
1.880
AGEB40-44 years                              2.15077    1.02567
2.097
AGEB45-49 years                              2.28321    1.02425
2.229
AGEB50-54 years                              2.56063    1.02335
2.502
AGEB55-59 years                              2.39072    1.02381
2.335
AGEB60-64 years                              2.59289    1.02289
2.535
AGEB65-69 years                              2.50395    1.02433
2.444
AGEB70-74 years                              2.29400    1.02801
2.232
AGEB75-79 years                              2.33331    1.03143
2.262
AGEB80-84 years                              2.23343    1.03805
2.152
AGEB85 years and older                       2.33522    1.05602
2.211
SEXMale                                      0.05957    0.07194
0.828
WAISTNot at risk                            -0.55838    0.11470
-4.868
WAISTSubstantially increased risk            0.47850    0.08948
5.348
DSMKHHRBOne                                  0.27778    0.09755
2.848
```

```
DSMKHHRBTwo                                0.58247    0.17380
3.351
DSMKHHRBThree or more                      0.60324    0.59302
1.017

Intercepts:
                Value   Std. Error t value
Low|Borderline   4.1233  1.0228     4.0313
Borderline|High  4.5601  1.0231     4.4572

Residual Deviance: 6144.936
AIC: 6190.936
(10065 observations deleted due to missingness)
```

It seems a lot more stable here, since there are no warnings this time around. In fact, we can do `summary()` on the `polr` objects, where the previous case we couldn't for some reason, likely due to problems with the Hessian.

Thus, the bottom line is that we may consider these models, but ONLY using `fdat.full`, as I don't think the model is meaningful if it is unstable or doesn't converge.

Assumptions Checking

The key assumptions of Ordinal Logistic Regression are:

1. Ordinal dependent variable
2. Independent variable types
3. Independence of assumptions
4. No multicollinearity
5. Proportional odds assumption (POA)

Assumptions I-III

The first assumption is satisfied based on how we defined `ALT3` and `GGT3` (and by extension `ALTRESB` and `GGTRESB`), and we have made that explicitly clear by mutating our data sets. The second and third assumptions are also held according to the NHS data set and variables we used.

Assumption IV

The fourth assumption can be checked using the VIF (variance inflation factor); we check if these values are less than 5 or 10. We use the

`performance::check_collinearity()` function for the `clm` models, and `car::vif` for the `polr` models.

▼ Code

```
# VIF Computation for models using full data set
check_collinearity(alt.folr)
```

```
# Check for Multicollinearity

Low Correlation

     Term  VIF   VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
 EXREGUIC 1.05 [1.02, 1.09]     1.02      0.96     [0.92, 0.98]
    AGEB 1.14 [1.11, 1.18]     1.07      0.88     [0.85, 0.90]
      SEX 1.03 [1.01, 1.08]     1.02      0.97     [0.93, 0.99]
   WAIST 1.14 [1.11, 1.18]     1.07      0.88     [0.85, 0.90]
 DSMKHHRB 1.04 [1.02, 1.09]     1.02      0.96     [0.92, 0.98]
```

▼ Code

```
check_collinearity(ggt.folr)
```

```
# Check for Multicollinearity

Low Correlation

     Term  VIF   VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
 EXREGUIC 1.04 [1.02, 1.09]     1.02      0.96     [0.92, 0.98]
    AGEB 1.11 [1.08, 1.15]     1.05      0.90     [0.87, 0.92]
      SEX 1.02 [1.00, 1.09]     1.01      0.98     [0.92, 1.00]
   WAIST 1.07 [1.05, 1.11]     1.04      0.93     [0.90, 0.95]
 DSMKHHRB 1.07 [1.04, 1.10]     1.03      0.94     [0.91, 0.96]
```

▼ Code

```
vif(alt.fpolr)
```

```
            GVIF Df GVIF^(1/(2*Df))
EXREGUIC 1.045963  1        1.022723
AGEB     1.137247 14        1.004604
SEX      1.034092  1        1.016903
WAIST    1.139798  2        1.033254
DSMKHHRB 1.044552  3        1.007291
```

▼ Code

```
vif(ggt.fpolr)
```

```
            GVIF Df GVIF^(1/(2*Df))
EXREGUIC 1.044362  1        1.021941
AGEB     1.112777 14        1.003824
SEX      1.016887  1        1.008408
WAIST    1.072834  2        1.017731
DSMKHHRB 1.065110  3        1.010568
```

In this case, we may want to keep all the models for now.

Assumption V

We will use the `brant` package for the Brant test for our `polr` models, and `nominal_test` for our `clm` models.

▼ Code

```
# Brant test for polr models
brant(alt.fpolr)
```

```
-------------------------------------------------------------------
--------------------
Test for                        X2  df  probability
-------------------------------------------------------------------
--------------------
Omnibus                         57.01   21  0
EXREGUICDid not meet recommended guidelines 3.7 1   0.05
AGEB20-24 years                 0.01    1   0.94
AGEB25-29 years                 0.98    1   0.32
AGEB30-34 years                 0.66    1   0.42
AGEB35-39 years                 0.01    1   0.93
AGEB40-44 years                 0.18    1   0.67
AGEB45-49 years                 0.77    1   0.38
AGEB50-54 years                 1.01    1   0.31
AGEB55-59 years                 0.46    1   0.5
AGEB60-64 years                 1.73    1   0.19
AGEB65-69 years                 0.42    1   0.52
AGEB70-74 years                 1.78    1   0.18
AGEB75-79 years                 0.02    1   0.88
AGEB80-84 years                 3.24    1   0.07
AGEB85 years and older              0.03    1   0.86
SEXMale                         14.77   1   0
WAISTNot at risk                    0.92    1   0.34
WAISTSubstantially increased risk           0.99    1   0.32
```

```
DSMKHHRBOne                        0.37    1   0.54
DSMKHHRBTwo                        2.57    1   0.11
DSMKHHRBThree or more                      0.36    1   0.55
-------------------------------------------------------------------
---------------------

H0: Parallel Regression Assumption holds
```

▼ Code

```
brant(ggt.fpolr)
```

```
-------------------------------------------------------------------
---------------------
Test for                           X2  df  probability
-------------------------------------------------------------------
---------------------
Omnibus                            -639.57 21  1
EXREGUICDid not meet recommended guidelines 0.84    1   0.36
AGEB20-24 years                    7.19    1   0.01
AGEB25-29 years                    17.6    1   0
AGEB30-34 years                    18.21   1   0
AGEB35-39 years                    23.99   1   0
AGEB40-44 years                    29.5    1   0
AGEB45-49 years                    40.71   1   0
AGEB50-54 years                    47.26   1   0
AGEB55-59 years                    41.83   1   0
AGEB60-64 years                    52.92   1   0
AGEB65-69 years                    31.58   1   0
AGEB70-74 years                    22.01   1   0
AGEB75-79 years                    12.49   1   0
AGEB80-84 years                    12.9    1   0
AGEB85 years and older                 3.22    1   0.07
SEXMale                            6.37    1   0.01
WAISTNot at risk                       1.45    1   0.23
WAISTSubstantially increased risk          0.72    1   0.4
DSMKHHRBOne                        3.51    1   0.06
DSMKHHRBTwo                        0   1   0.99
DSMKHHRBThree or more                      1155.9  1   0
-------------------------------------------------------------------
---------------------

H0: Parallel Regression Assumption holds
```

▼ Code

```
# Nominal test for clm models
```

```
nominal_test(alt.folr)
```

```
Tests of nominal effects

formula: ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB
         Df  logLik    AIC     LRT  Pr(>Chi)
<none>      -3501.0 7047.9
EXREGUIC  1 -3497.7 7043.4  6.5275   0.01062 *
AGEB     14 -3488.9 7051.9 24.0724   0.04491 *
SEX       1 -3491.1 7030.2 19.7755 8.709e-06 ***
WAIST     2 -3498.2 7046.3  5.5944   0.06098 .
DSMKHHRB  3 -3498.7 7049.5  4.4730   0.21472
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

▼ Code

```
nominal_test(ggt.folr)
```

```
Tests of nominal effects

formula: GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB
         Df  logLik    AIC    LRT Pr(>Chi)
<none>      -3072.5 6190.9
EXREGUIC  1 -3071.9 6191.7 1.2156  0.27023
AGEB
SEX       1 -3069.5 6187.0 5.9221  0.01495 *
WAIST     2 -3071.6 6193.2 1.7086  0.42557
DSMKHHRB
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

For the Brant tests, we got the warning: "1351 combinations in table(dv,ivs) do not occur. Because of that, the test results might be invalid.". So, I'm not sure if i can really do much inference from that.

Additionally, since we want the p-values $\Pr(> \text{Chi})$ to not be significant for our nominal tests, it seems that the `clm` models do not satisfy this assumption either.

Takeaway

Our ordinal logistic regression models violate the POA, which tells us that **the risks for these health markers do not change uniformly as we move from Low $\rightarrow$ Borderline $\rightarrow$ High.**

- In both our ALT and GGT models, the variable `SEX` rejected the proportional

odds assumption. It shows that being male or female impacts the shift from Low to Borderline completely differently than it impacts the shift from Borderline to High. A single, global slope coefficient for `SEX` would mask this reality.

- The proportional-odds assumption is not supported for all predictors, indicating that some covariates have different associations at different severity thresholds. Consequently, a multinomial model may be preferable for the primary analysis because it allows category-specific effects without imposing proportional odds.
- The broken Brant test math confirmed that certain age brackets combined with specific outcome categories have near-zero occurrences. This can tell us about how our demographics are and in general the natural clusterings. Overall, it also suggests collapsing `AGEB` even more at some point as well.

Continuation Ratio Models

These are ordinal logistic regression models specifically designed for cases when proportional odds are not satisfied. We relax the requirement that the coefficients must stay the same and instead have two pairs of coefficients, one for Low $\rightarrow$ Borderline or High, and the other for Borderline $\rightarrow$ High; this allows us to utilise the ordinal nature of our variables while not being completely restricted by POA. The implementation here is from `VGAM::vgam()`.

▼ Code

```
# Fitting CRM models on full data set
alt.crm = vgam(ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
               family = cratio(parallel = FALSE), data = fdat.fu
summary(alt.crm)
```

```
Call:
vgam(formula = ALT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
    family = cratio(parallel = FALSE), data = fdat.full)

Names of additive predictors: logitlink(P[Y>1|Y>=1]),
logitlink(P[Y>2|Y>=2])

Dispersion Parameter for cratio family:   1

Residual deviance:  6939.134 on 10776 degrees of freedom

Log-likelihood: -3469.567 on 10776 degrees of freedom
```

```
Number of Fisher scoring iterations:  7

DF for Terms and Approximate Chi-squares for Nonparametric
Effects

             Df Npar Df Npar Chisq P(Chi)
(Intercept):1  1
(Intercept):2  1
EXREGUIC:1     1
EXREGUIC:2     1
AGEB:1        14
AGEB:2        14
SEX:1          1
SEX:2          1
WAIST:1        2
WAIST:2        2
DSMKHHRB:1     3
DSMKHHRB:2     3
```

▼ Code

```
OR.alt = exp(coef(alt.crm))
OR.alt.CI = exp(confintvglm(alt.crm))

ggt.crm = vgam(GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
               family = cratio(parallel = FALSE), data = fdat.fu
summary(ggt.crm)
```

```
Call:
vgam(formula = GGT3 ~ EXREGUIC + AGEB + SEX + WAIST + DSMKHHRB,
    family = cratio(parallel = FALSE), data = fdat.full)

Names of additive predictors: logitlink(P[Y>1|Y>=1]),
logitlink(P[Y>2|Y>=2])

Dispersion Parameter for cratio family:   1

Residual deviance:  6115.568 on 10776 degrees of freedom

Log-likelihood: -3057.784 on 10776 degrees of freedom

Number of Fisher scoring iterations:  16

DF for Terms and Approximate Chi-squares for Nonparametric
Effects
```

```
             Df Npar Df Npar Chisq P(Chi)
(Intercept):1  1
(Intercept):2  1
EXREGUIC:1     1
EXREGUIC:2     1
AGEB:1        14
AGEB:2        14
SEX:1          1
SEX:2          1
WAIST:1        2
WAIST:2        2
DSMKHHRB:1     3
DSMKHHRB:2     3
```

▼ Code

```
OR.ggt = exp(coef(ggt.crm))
OR.ggt.CI = exp(confintvglm(ggt.crm))
```

▼ Code

```
# Table for OR.alt
or_df = data.frame(RawName = names(OR.alt), OR = as.numeric(OR.a
                   LowerCI = OR.alt.CI[, 1], UpperCI = OR.alt.CI
                   stringsAsFactors = FALSE)
or_df = or_df[!grepl("(Intercept)", or_df$RawName), ]
or_df$Variable = str_remove(or_df$RawName, ":[12]$")
or_df$Step = ifelse(str_detect(or_df$RawName, ":1$"),
                    "Step 1: Low vs Borderline/High",
                    "Step 2: Borderline vs High")
final_table = or_df[, c("Variable", "Step", "OR", "LowerCI", "Up
final_table[, c("OR", "LowerCI", "UpperCI")] =
  round(final_table[, c("OR", "LowerCI", "UpperCI")], 3)
rownames(final_table) = NULL
kable(final_table, caption = "CRM Model Odds Ratio for ALT",
      align = "llccc",
      col.names = c("Variable", "Step", "OR", "Lower 95% CI", "U
```

CRM Model Odds Ratio for ALT

Variable	Step	OR	Lower 95% CI	Upper 95% CI
EXREGUICDid not meet recommended guidelines	Step 1: Low vs Borderline/High	1.093	0.954	1.253
EXREGUICDid not meet	Step 2: Borderline vs	1.294	1.025	1.634

recommended guidelines	High			
AGEB20-24 years	Step 1: Low vs Borderline/High	1.044	0.451	2.413
AGEB20-24 years	Step 2: Borderline vs High	1.072	0.224	5.133
AGEB25-29 years	Step 1: Low vs Borderline/High	1.534	0.711	3.311
AGEB25-29 years	Step 2: Borderline vs High	0.619	0.149	2.574
AGEB30-34 years	Step 1: Low vs Borderline/High	1.232	0.570	2.662
AGEB30-34 years	Step 2: Borderline vs High	0.623	0.149	2.601
AGEB35-39 years	Step 1: Low vs Borderline/High	1.320	0.616	2.830
AGEB35-39 years	Step 2: Borderline vs High	1.259	0.304	5.212
AGEB40-44 years	Step 1: Low vs Borderline/High	1.185	0.556	2.528
AGEB40-44 years	Step 2: Borderline vs High	0.816	0.200	3.336
AGEB45-49 years	Step 1: Low vs Borderline/High	1.629	0.769	3.451
AGEB45-49 years	Step 2: Borderline vs High	0.684	0.170	2.752
AGEB50-54 years	Step 1: Low vs Borderline/High	1.672	0.790	3.541
AGEB50-54 years	Step 2: Borderline vs High	0.596	0.148	2.397
AGEB55-59 years	Step 1: Low vs Borderline/High	1.212	0.570	2.574
AGEB55-59 years	Step 2: Borderline vs High	0.675	0.166	2.735
AGEB60-64 years	Step 1: Low vs Borderline/High	1.316	0.622	2.785
AGEB60-64 years	Step 2: Borderline vs High	0.469	0.117	1.889
AGEB65-69 years	Step 1: Low vs Borderline/High	0.887	0.414	1.897

AGEB65-69 years	Step 2: Borderline vs High	0.581	0.142	2.389
AGEB70-74 years	Step 1: Low vs Borderline/High	0.665	0.304	1.453
AGEB70-74 years	Step 2: Borderline vs High	0.348	0.081	1.489
AGEB75-79 years	Step 1: Low vs Borderline/High	0.261	0.110	0.619
AGEB75-79 years	Step 2: Borderline vs High	0.729	0.143	3.712
AGEB80-84 years	Step 1: Low vs Borderline/High	0.195	0.074	0.517
AGEB80-84 years	Step 2: Borderline vs High	0.057	0.005	0.685
AGEB85 years and older	Step 1: Low vs Borderline/High	0.066	0.013	0.321
AGEB85 years and older	Step 2: Borderline vs High	0.502	0.022	11.275
SEXMale	Step 1: Low vs Borderline/High	2.473	2.157	2.835
SEXMale	Step 2: Borderline vs High	0.738	0.583	0.934
WAISTNot at risk	Step 1: Low vs Borderline/High	0.515	0.420	0.633
WAISTNot at risk	Step 2: Borderline vs High	0.700	0.481	1.020
WAISTSubstantially increased risk	Step 1: Low vs Borderline/High	1.900	1.605	2.249
WAISTSubstantially increased risk	Step 2: Borderline vs High	1.402	1.049	1.872
DSMKHHRBOne	Step 1: Low vs Borderline/High	0.833	0.687	1.011
DSMKHHRBOne	Step 2: Borderline vs High	1.108	0.792	1.549
DSMKHHRBTwo	Step 1: Low vs Borderline/High	0.726	0.504	1.047
DSMKHHRBTwo	Step 2: Borderline vs High	1.514	0.777	2.950
DSMKHHRBThree or more	Step 1: Low vs	1.299	0.419	4.027

	Borderline/High			
DSMKHHRBThree or more	Step 2: Borderline vs High	0.537	0.086	3.363

▼ Code

```
# Table for OR.ggt
or_df2 = data.frame(RawName = names(OR.alt), OR = as.numeric(OR.
                    LowerCI = OR.alt.CI[, 1], UpperCI = OR.alt.C
                    stringsAsFactors = FALSE)
or_df2 = or_df2[!grepl("(Intercept)", or_df$RawName), ]
or_df2$Variable = str_remove(or_df2$RawName, ":[12]$")
or_df2$Step = ifelse(str_detect(or_df2$RawName, ":1$"),
                     "Step 1: Low vs Borderline/High",
                     "Step 2: Borderline vs High")
final_table2 = or_df2[, c("Variable", "Step", "OR", "LowerCI", "
final_table2[, c("OR", "LowerCI", "UpperCI")] =
  round(final_table2[, c("OR", "LowerCI", "UpperCI")], 3)
rownames(final_table2) = NULL
kable(final_table2, caption = "CRM Model Odds Ratio for GGT",
      align = "llccc",
      col.names = c("Variable", "Step", "OR", "Lower 95% CI", "U
```

CRM Model Odds Ratio for GGT

Variable	Step	OR	Lower 95% CI	Upper 95% CI
(Intercept)	Step 1: Low vs Borderline/High	0.016	0.067	0.300
(Intercept)	Step 2: Borderline vs High	42845.887	0.375	6.112
EXREGUICDid not meet recommended guidelines	Step 1: Low vs Borderline/High	1.352	0.954	1.253
EXREGUICDid not meet recommended guidelines	Step 2: Borderline vs High	1.291	1.025	1.634
AGEB20-24 years	Step 1: Low vs Borderline/High	4.464	0.451	2.413
AGEB20-24 years	Step 2: Borderline vs High	0.000	0.224	5.133
AGEB25-29 years	Step 1: Low vs Borderline/High	5.626	0.711	3.311
AGEB25-29 years	Step 2: Borderline vs High	0.000	0.149	2.574

AGEB30-34 years	Step 1: Low vs Borderline/High	7.162	0.570	2.662
AGEB30-34 years	Step 2: Borderline vs High	0.000	0.149	2.601
AGEB35-39 years	Step 1: Low vs Borderline/High	7.054	0.616	2.830
AGEB35-39 years	Step 2: Borderline vs High	0.000	0.304	5.212
AGEB40-44 years	Step 1: Low vs Borderline/High	8.625	0.556	2.528
AGEB40-44 years	Step 2: Borderline vs High	0.000	0.200	3.336
AGEB45-49 years	Step 1: Low vs Borderline/High	9.962	0.769	3.451
AGEB45-49 years	Step 2: Borderline vs High	0.000	0.170	2.752
AGEB50-54 years	Step 1: Low vs Borderline/High	12.994	0.790	3.541
AGEB50-54 years	Step 2: Borderline vs High	0.000	0.148	2.397
AGEB55-59 years	Step 1: Low vs Borderline/High	11.029	0.570	2.574
AGEB55-59 years	Step 2: Borderline vs High	0.000	0.166	2.735
AGEB60-64 years	Step 1: Low vs Borderline/High	13.365	0.622	2.785
AGEB60-64 years	Step 2: Borderline vs High	0.000	0.117	1.889
AGEB65-69 years	Step 1: Low vs Borderline/High	12.107	0.414	1.897
AGEB65-69 years	Step 2: Borderline vs High	0.000	0.142	2.389
AGEB70-74 years	Step 1: Low vs Borderline/High	9.961	0.304	1.453
AGEB70-74 years	Step 2: Borderline vs High	0.000	0.081	1.489
AGEB75-79 years	Step 1: Low vs Borderline/High	10.219	0.110	0.619
AGEB75-79 years	Step 2: Borderline vs	0.000	0.143	3.712

	High			
AGEB80-84 years	Step 1: Low vs Borderline/High	9.623	0.074	0.517
AGEB80-84 years	Step 2: Borderline vs High	0.000	0.005	0.685
AGEB85 years and older	Step 1: Low vs Borderline/High	10.160	0.013	0.321
AGEB85 years and older	Step 2: Borderline vs High	0.000	0.022	11.275
SEXMale	Step 1: Low vs Borderline/High	1.083	2.157	2.835
SEXMale	Step 2: Borderline vs High	0.740	0.583	0.934
WAISTNot at risk	Step 1: Low vs Borderline/High	0.566	0.420	0.633
WAISTNot at risk	Step 2: Borderline vs High	1.178	0.481	1.020
WAISTSubstantially increased risk	Step 1: Low vs Borderline/High	1.609	1.605	2.249
WAISTSubstantially increased risk	Step 2: Borderline vs High	1.338	1.049	1.872
DSMKHHRBOne	Step 1: Low vs Borderline/High	1.358	0.687	1.011
DSMKHHRBOne	Step 2: Borderline vs High	0.773	0.792	1.549
DSMKHHRBTwo	Step 1: Low vs Borderline/High	1.787	0.504	1.047
DSMKHHRBTwo	Step 2: Borderline vs High	1.142	0.777	2.950
DSMKHHRBThree or more	Step 1: Low vs Borderline/High	1.631	0.419	4.027
DSMKHHRBThree or more	Step 2: Borderline vs High	1943957.549	0.086	3.363

Not quite sure why some of the point estimates lie outside the confidence intervals here; it could be some bug in my code. I may fix it at some point but as it stands, we will probably just stick with Multinomial Logistic Regression anyway so it does not really matter. Another model that we can look into at some point is the Partial Proportional Odd Model, which also relaxes the POA. This is implemented in `VGAM::vglm/vgam` with `family=cumulative(parallel = FALSE ~ x1)`.